

Study workflow

Dataset 1, Urinalysis CKD
380 patients, 20 predictors
239 CKD and 141 non CKD



Preprocessing inside each fold: categorical encoding, median and mode imputation, scaling for logistic regression, SVM, and MLP, class weighting



Stratified 5 by 3 nested cross validation



8 models: logistic regression, decision tree, random forest, XGBoost, LightGBM, CatBoost, SVM, and MLP



Evaluation: ROC AUC, PR AUC, F1, MCC, sensitivity, specificity, balanced accuracy, Brier, threshold by MCC, bootstrap confidence intervals



Model selection, SHAP stability, permutation importance, PDP and ICE, local SHAP and LIME, learning curve